This article was downloaded by: [Ondokuz Mayis Universitesine]
On: 08 November 2014, At: 13:41
Publisher: Taylor & Francis
Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Journal of Immunoassay and Immunochemistry

Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/ljii20

Seroprevalence Rates of Hepatitis C Virus (Hcv) Antibody And Hepatitis B Virus Surface Antigen (Hbsag) in Blood Donors in A Southwestern Nigerian City

Babatunde O. Motayo[ab], Onoja M. Akpa[c], Isreal Ezeani[d], Adedayo O. Faneye[b], Usen A. Udo[d] & Bernard Onoja[b]

[a] Microbiology Unit, Pathology Department, Federal Medical Centre, Abeokuta, Nigeria

[b] Department of Virology, College of Medicine, University of Ibadan, Nigeria

[c] Department of Epidemiology and Medical Statistics, College of Medicine, University of Ibadan, Nigeria

[d] Blood Transfusion Unit, Pathology Department, Federal Medical Centre, Abeokuta, Nigeria

Accepted author version posted online: 14 Mar 2014.Published online: 22 Aug 2014.

To cite this article: Babatunde O. Motayo, Onoja M. Akpa, Isreal Ezeani, Adedayo O. Faneye, Usen A. Udo & Bernard Onoja (2015) Seroprevalence Rates of Hepatitis C Virus (Hcv) Antibody And Hepatitis B Virus Surface Antigen (Hbsag) in Blood Donors in A Southwestern Nigerian City, Journal of Immunoassay and Immunochemistry, 36:1, 91-99, DOI: 10.1080/15321819.2014.899256

To link to this article: http://dx.doi.org/10.1080/15321819.2014.899256

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or

howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at http://www.tandfonline.com/page/terms-and-conditions

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

Journal of Immunoassay and Immunochemistry, 36:91–99, 2015
Copyright © Taylor & Francis Group, LLC
ISSN: 1532-1819 print/1532-4230 online
DOI: 10.1080/15321819.2014.899256

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

SEROPREVALENCE RATES OF HEPATITIS C VIRUS (HCV) ANTIBODY AND HEPATITIS B VIRUS SURFACE ANTIGEN (HBsAg) IN BLOOD DONORS IN A SOUTHWESTERN NIGERIAN CITY

Babatunde O. Motayo,[1,2] Onoja M. Akpa,[3] Isreal Ezeani,[4] Adedayo O. Faneye,[2] Usen A. Udo,[4] and Bernard Onoja[2]

[1] *Microbiology Unit, Pathology Department, Federal Medical Centre, Abeokuta, Nigeria*
[2] *Department of Virology, College of Medicine, University of Ibadan, Nigeria*
[3] *Department of Epidemiology and Medical Statistics, College of Medicine, University of Ibadan, Nigeria*
[4] *Blood Transfusion Unit, Pathology Department, Federal Medical Centre, Abeokuta, Nigeria*

□ *Hepatitis B and C have been identified as major causes of Transfusion transmitted infections, in Nigeria. Our objective was to determine the prevalence of Hepatitis B virus (HBV) and Hepatitis C virus (HCV) in prospective blood donors in Abeokuta, Nigeria. 305 blood donors were screened for the presence of Hepatitis B virus surface antigen (HBsAg) and HCV using a rapid immunochromatographic kit (DiaSpot®). Demographic information was also collected. Males constituted 96.4%, singles were the majority with 65%. Prevalence of HBsAg was 9.8%, HCV 1.3%, and dual positivity 0.3%. Prevalence of HBsAg and HCV among males was 10.2% and 1.4%, while females recorded 0.0% for HCV and HBsAg. Dual positivity was recorded in a male (0.33%). Analysis of the study variables revealed that only educational status was statistically associated with positivity of HBsAg* ($\chi^2 = 7.49,\ p = 0.02$), *HCV prevalence was highest in the illiterate group it was not statistically significant* ($P > 0.05$). *We report the prevalence rates of anti-HCV Ab and HBsAg in blood donors from Ogun State, Nigeria. Our results reveals higher rate of HBsAg and evidence of co-infection with both viruses, illiteracy was the only variable associated with HBV infection. We advocate for the inclusion of anti-HBc or HBeAg in donor screening in our environment.*

Keywords HBsAg, HCV, seoprevailence, blood donors, abeokuta

INTRODUCTION

Viral Hepatitis particularly HBV and HCV have been recognized as major blood borne infectious diseases in sub-Saharan Africa, with 31.9 million HBV

Address correspondence to Babatunde O. Motayo, Microbiology Unit, Pathology Department, Federal Medical Centre, Abeokuta. P.M.B. 3031 Sapon Abeokuta, Nigeria. E-mail: babatundemotayo@yahoo.com

Color versions of one or more of the figures in this article can be found online at www.tandfonline.com/ljii.

infections occurring annually.[1] Surveillance data has shown that up to 15–60% of the general population in Africa may be positive to one or more serological markers of HBV.[2] Blood transfusion has further increased the global burden of these diseases. As a result, government and agencies have adopted various policies and programs directed towards increased blood safety for transfusion. In most developed countries, more reliable but expensive diagnostic screening methods are being used in order to reduce the rate of TTI's. In Sub-Saharan Africa, however, availability of these technologies remain elusive because of the high cost. Previous research has shown the risk of exposure by patients to infection with HBV through occult infection with HBV in HBsAg negative blood donors, in Ife, Nigeria.[3]

Infection by either HBV or HCV is first recognized by non specific immune response before specific cellular and then humoral immune responses.[4–7] The virus-infected cells are detected by cytotoxic CD8+ cells that bind to endogenously processed viral antigens expressed mostly on the surface of hepatocytes, leading to lysis and most often immune mediated liver damage.[8] Immunocompetent persons often recover following virus clearance and production of neutralizing antibodies which confers life-long immunity.[9,10] In some cases, however, primary infection with either HBV or HCV can lead to chronic disease if medical attention is not given promptly, which will lead to liver cirrhosis or hepatocellular carcinoma (HCC).[8] Various viral as well as host factors have been investigated for development of persistent infections, with dual infection, either co-infection or super infection been a major risk factor for development of Hepatocellular carcinoma (HCC).[11,12] Certain risk factors have been identified with co-infection with HBV and HCV this include, injection drug usage (IDU), blood and organ recipients, HIV infection, certain ethnicities and races.[7,13]

In Nigeria, several reports have shown the presence of HBV and HCV antibodies in healthy individuals, as well as voluntary blood donors.[8,14,15] With the increase in incidence of HCC in Nigerians with history of viral hepatitis,[16] and evidence of possible post-transfusion HBV infection, it is pertinent to continue to investigate the prevalence of HBV and HCV infections in prospective blood donors. Particularly when the National Blood Transfusion Service (NBTS) is advocating the use of more sensitive blood screening systems such as 3rd generation ELISA. The present research prospectively studied the prevalence of mono and dual HBsAg and HCV antibodies in voluntary blood donors in a Nigerian City.

MATERIALS AND METHODS

Study Design and Study Population

The current study is a prospective crossectional study of voluntary intending blood donors. The study was designed as part of the hospital

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

blood safety surveillance program in line with the mandate of the Federal Ministry of health, national blood transfusion program (NBTS) and Safe Blood for Africa. Ethical approval was sought and given. Consecutive sampling was adopted in recruiting participants for the study. Study duration was from March 2013 to June 2013. Abeokuta is the capital city of Ogun State, southwest, Nigeria. The study site was the blood transfusion unit of the Federal Medical Centre, Abeokuta, which serves the blood transfusion needs of patients in Ogun State and its environs with about 3.5 million inhabitants.[17]

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

Donor Recruitment

Participants for this study were drawn from replacement donors as well as voluntary blood donors consenting to partake in the study. Participants were counseled before filling the donor/consent from. Their blood pressure was taken, prior to sample collection and preliminary screening. Tests done include blood group, and screening for TTI, HIV 1&2, HBV surface Ag, HCV, and *Treponema pallidium* antibodies (VDRL) following NBTS recommended algorithm. In our center, blood test results are not released to donors.

Serologic Testing

Blood samples drawn into anticoagulant EDTA bottles were centrifuged at low speed for separation of plasma. Two serologic test were performed simultaneously for the purpose of this study, they are HBsAg, using a commercial rapid EIA based kit DiaSpot® (Bresta Perkasa, Djarkarta, Indonesia). This combines a lateral flow system and an immunochromatographic assay, (specificity 97%, sensitivity 99%, respectively). The anti-HCV test was carried using a similar test kit DiaSpot (Bresta, Perkasa, Jakarta, Indonesia), which uses the same test principle as that of HBsAg. Test protocol was done according to the manufacturer's instructions.

Statistical Analysis

Demographic and laboratory data generated were grouped averaged and analyzed using SPSS vs. 15 for windows. Associations or "a lack thereof" between variables and prevalence rates of mono or dual infections were determined using Chi square test. A p-value of 0.05 or less was set as significant for analyzed variables.

RESULTS

Sociodemographic Characteristics and Prevalence Rates of HBsAg and anti-HCV among Blood Donors

In Table 1, majority of the blood donors were males (96.4%) while only 3.6% of them are female. Also, while 46.2% of them were between 30 to 40 years old, 36.4% of them were less than 30 years old. Although most of them (42.0%) had completed up to secondary education, 16.4% of them were students while 3.6% and 30.5% of them were employees of private organizations' and the government respectively. In addition to that, the overall prevalence rates of HBsAg and anti-HCV among the donors were 9.8% and 1.3%, respectively (Table 1).

TABLE 1 Socio-demographic characteristics and serologic prevalence rates among intending blood donors in FMC, Abeokuta, 2013

Variable	Frequency	Percentage
Sex		
Female	11	3.6
Male	294	96.4
Age		
<30	111	36.4
30-40	141	46.2
>40	48	15.7
Not reported	5	1.6
Highest level of education		
No formal education	42	13.8
Up to secondary school	128	42.0
Graduate	115	37.7
Marital Status		
Married	3	1.0
Single	199	65.2
Occupation		
Student	50	16.4
Self employed	73	23.9
Private employee	11	3.6
Government employee	93	30.5
Others	43	14.1
Not reported	35	11.5
HBsAg diagnosis		
Positive	30	9.8
Negative	275	90.2
HCV diagnosis		
Positive	4	1.3
Negative	301	98.7

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

Serologic Results of Donors Plasma Samples

In Table 2, prevalence rates of HBsAg and HCV among male donors were 10.2% and 1.4%, respectively, while prevalence rates of HBsAg and HCV among female donors was 0.0% each. The only participant recorded "dual positivity (0.33%)" is a male. Also, while the prevalence rates of HBsAg, HCV and dual infection were 21.4%, 2.4%, and 2.4% among participants with no formal education, the prevalence of HBsAg was 7.8% among donors with up to secondary education and those with tertiary education. However, among participants with up to secondary education and those with tertiary education, prevalence rates of anti-HCV antibodies were 1.6 and 0.9%, respectively.

Furthermore, while prevalence rate of HBsAg was higher among participants that were married (10.7), prevalence rate of HCV was higher among participants that were not married (1.5%) (Table 2). In addition, when comparing dual infection for HBsAg and HCV (Tables 1 and 2), 9.8% of intending donors were positive for HBsAg and 98.7% were negative for anti-HCV while only 0.33% were positive for both HBsAg and anti-HCV antibodies.

TABLE 2 Prevalence rates of single and dual positivity of HBsAg and anti-HCV antibody among intending blood donors in FMC, Abeokuta, 2013

	HBsAg			HCV			
Variable	Positive	Negative	χ^2	Positive	Negative	χ^2	Dual Infection
Sex							
Male	30(10.2)	264(89.8)	1.25	4(1.4)	290(98.6)	0.15	1(0.3)
Female	0(0.0)	11(100.0)		0(0.0)	11(100.0)		0(0.0)
Age							
<30	10(9.0)	101(91.0)	5.36	1(0.9)	110(99.1)	1.48	0(0.0)
30-40	19(13.5)	122(86.5)		3(2.1)	138(97.9)		1(0.7)
>40	1(2.1)	47(97.9)		0(0.0)	48(100.0)		0(0.0)
Highest level of education							
No formal education	9(21.4)	33(78.6)	7.48*	1(2.4)	41(97.6)	0.55	1(2.4)
Up to secondary school	10(7.8)	118(92.2)		2(1.6)	126(98.4)		0(0.0)
Graduate	9(7.8)	106(92.2)		1(0.9)	114(99.1)		0(0.0)
Marital Status							
Married	11(10.7)	92(89.3)	0.10	1(1.0)	102(99.0)	0.15	0(0.0)
Single	19(9.5)	180(90.5)		3(1.5)	196(98.5)		1(0.5)
Occupation							
Student	3(6.0)	47(94.0)	4.64	1(2.0)	49(98.0)	1.80	0(0.0)
Self employed	9(12.3)	64(87.7)		2(2.7)	71(97.3)		1(1.3)
Private employee	1(9.1)	10(90.9)		0(0.0)	11(100.0)		0(0.0)
Government employee	6(6.5)	87(93.5)		1(1.1)	92(98.9)		0(0.0)
Others	7(16.3)	36(83.7)		0(0.0)	43(100.0)		0(0.0)

*$p < 0.05$

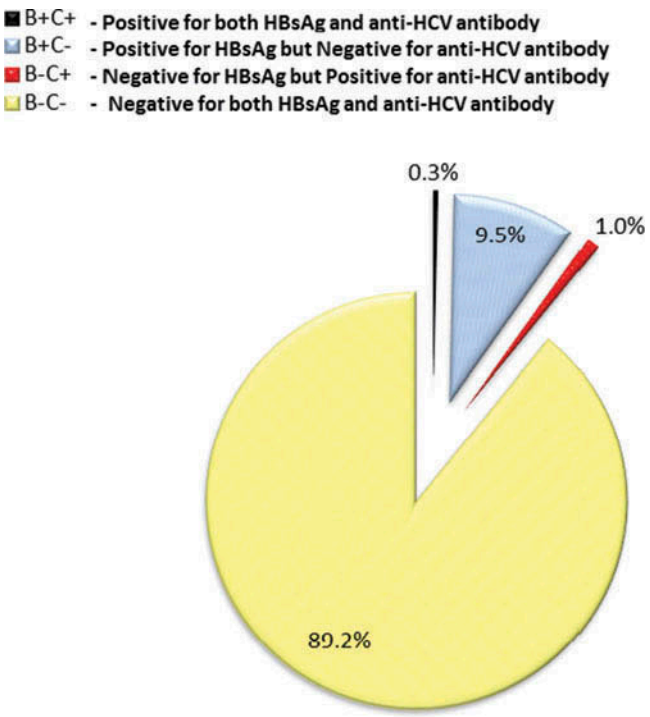


FIGURE 1 Proportion positive when comparing both HBsAg and anti-HCV antibody prevalence.

In Table 2, HBsAg was found to be associated with level of education ($\chi^2 = 7.49$, $p = 0.02$); in particular, the intending blood donors with no formal education who had group-specific prevalence rate of 21.4% were more likely to be HBsAg positive compared to either of the two other education categories.

In addition, when comparing positivity for HBsAg and anti-HCV antibody in the overall sample (Figure 1), 0.3% of the donors were positive for both HBsAg and anti-HCV antibody. However, while 9.5% of the donors were positive for HBsAg but negative for anti-HCV antibody, 1.0% of the donors were negative for HBsAg but positive for anti-HCV antibody.

DISCUSSION

Our current study aims at estimating the prevalence of mono and dual positivity of HBsAg and HCV amongst apparently healthy, blood donors in Abeokuta capital city of Ogun State, Nigeria. Prevalence of HBsAg was 30(9.8%), while anti-HCV antibodies (ab) recorded 4(1.31%) in our study population (n = 305), dual positivity with both viruses was recorded in only 1 participant (0.3%). Our result is consistent with a similar study which

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

reported a lower HCV infection rate as compared to HBV.[18] This can be naturally expected as various studies have shown the high rate infectiousness and resistance to inactivation by adverse environmental conditions.[19,20] However our findings is deviant to a recent study by Oje et al.[8] which reported a higher prevalence rate of HCV (12.8%) when compared to HBsAg (9.8%), even though the study in question was not carried out among blood donors it shows the possibility of variations from normally expected values. Most surveillance studies are based on members of the general population as these are more representative of age and sex; blood donors are often selected based good health and serve as a suitable population group when apparently healthy individuals are to be taken into consideration. This is particularly useful when rate of new infections are to be considered as is the case with our study. That is why a dual seroprevalence rate of 0.3% and 1.4% for anti-HCV antibody is indicative of a reduced transmission rate of HCV in our study setting when compared with other studies within the same geographical region as ours.[2,8] This shows an improvement in health promotion practices in Abeokuta, although this is yet to be justified and leaves a gap in knowledge for further study.

Demographic analysis of associated variables, revealed that males recorded a higher prevalence rate for both serological markers (10.2% for HBsAg, 1.4% anti-HCV antibodies), while females recorded zero prevalence rate for both HBsAg and anti-HCVab. This is expected as only 11 females were recruited in our study, however most studies with a fairly representative female population have reported higher prevalence rate of HBsAg and HCV in males than females.[8,14,21] This observed trend has been attributed to a higher risk of exposure and active lifestyle of most males to females (1). Age range distribution revealed that age 30<40 had the highest value for both HBsAg and anti-HCV, there was no significant association between age and seropostivity for both viruses (HBsAg, Chi square = 5.36, $p > 0.05$, HCV Chi square = 1.48, $p > 0.05$), our results are in consonant with some other reports from Nigeria, that show an upward trend in HBV/HCV prevalence rate around the age range of 30–45.[2,8] Analyzed data did not reveal any statistical relationship with regards to type of occupation and HBsAg or anti-HCVab seroprevalence, although, the self employed group recorded highest rate for both serological markers, 12.3% for HBsAg, and 4.7% anti-HCV antibody. This finding is similar to the report of Oje et al.,[8] which also reported that mono infection to HBV and HCV was highest in Artisans and Traders, majority of which are usually self employed. The reason for this observation was not independently investigated in our study, but could be ascertained to increased risk of exposure from cuts and bruises sustained during hard labor with sharp working tools. Marital status and HBsAg prevalence rate were not significantly associated ($p > 0.05$), married participants,

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

however, recorded a higher prevalence rate 10.7%, anti-HCVab, and singles displayed a higher prevalence rate (Table 2). Dual positivity to both virus markers was found among singles. A significant limitation to this study was our inability to consistently obtain data for number of sexual partners and condom usage as this would have shed more light on the relationship between sexual lifestyle preferences and risk of infection, this is however regretted. But from previous studies condom usage has shown significant reduction in infection rate of HBV/HCV as they are both sexually transmitted infections.[8]. Educational status was significantly associated with HBsAg positivity (Chi square 7.48, $p = 0.02$), this was the only variable that showed significant association in our current study, though anti-HCV prevalence rate was highest in the informal education group (2.4%) about 3 times the value for the graduate group (Table 2). This observed trend confirms that literacy contributes significantly to improved personal health promotion practices as previous reported by Oje et al.,[8] although lifestyle changes have been reported to be more effective in the control of transmission of sexually transmitted infections such as HIV and HBV.[6,22]

CONCLUSION

In conclusion, we report the prevalence of mono and dual positivity of anti-HCV antibody and HBsAg in blood donors from Ogun State, southwest, Nigeria for the first time. Statistical evidence from our study reveal that illiteracy remains one of the driving forces for these infections in resource poor settings such as ours. We advocate that more effort be channeled into mass mobilization against blood borne infections such as HBV, and HCV through preferably the electronic media, in form of jingles and short messages in the local languages spoken in the region. Health authorities particularly the NBTS, Federal ministry of Health, Nigeria, should as a matter of urgency consider the inclusion of a second screening test for HBV preferably the anti-HBc or HBeAg test for HBsAg negative prospective blood donors, to prevent accidental transmission of HBV by occult HBV blood donors, the presence of which has been previously reported in Nigeria.[3,16]

REFERENCES

1. Madhara, V.; Burgess, C.; Prucker E. Epidemiology of Chronic HCV in Sub-Saharan, Africa. *Lancet* **2002**, *2157*, 293–302.
2. Balogun, M.T.; Emmanuel, S.; Ojerinde, E.F. HIV, Hepatitis B and C Viruses Co-infection among Patients in a Nigerian Tertiary Hospital. *Pan. Afr. Med. J*. **2012**, *12*,100.
3. Japhet, M.O.; Adesina, O.A.; Donbraye, E.; Adewumi, O.M. Hepatitis B Core IgM Antibody among HBAg Negative Blood Donors in Nigeria. *Virol. J*. **2011**, *8*, 513.
4. Bigger, C.B.; Brasky, K.M.; Landford, R.E. DNA Micro-Array of Chimpanzee Liver during Acute Resolving Hepatitis C Infection. *J. Virol*. **2001**, *75*, 7059–7066.

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014

5. Thimme, R.; Bukh, J.; Spangenberg, H.C.; Wieland, S.; Pemberton, J.; Steiger, C.; Govindarajan, S.; Purcell, R.H.; Chisar, F.V. Viral and Immunological Determinants of HCV Clearance, Persistence and Disease. *Proc. Natl. Acad. Sci. USA*. **2002**, *99*, 15661–15668.
6. Fayemiwo, S.A.; Odiabo, G.N.; Oni, A.A.; Ajayi, A.A.; Bakare, R.A.; Olaleye, D.O. Genital Ulcer Disease among HIV- Infected Commercial Sex Workers in Ibadan, Nigeria. *Afr. J. Med. Med. Sci*. **2011**, *40*(1), 39–46.
7. Su, A.I.; Pezacki, J.P.; Woodika, L.; Brideau, A.D.; Supekova, L.; Thimme, R.; Weiland, S.; Bukh, J.; Prucell, R.H.; Schultz, P.G.; Chisari, F.V. Genomic Analysis of the Host Response to Hepatitis C Infection. *Proc. Natl. Acad. Sci. USA* **2002**, *99*, 15669–15674.
8. Oje, O.J.; Sule, W.F.; Famurewa, D. Dual Positivity of Hepatitis B Surface Antigen and Anti- Hepatitis C Virus Antibody and Associated Risk Factors among Apparently Healthy Patients of Ekiti State, Nigeria. *Viral Immunol*. **2012**, *25*(6), 448–455.
9. Strader, D.B.; Wright, T.; Thoma, D.L.; Sheff, L.B. Diagnosis, Management and Treatment of Hepatitis C. *Hepatology* **2004**, *39*, 1147–1171.
10. Locarnini, S. Molecular Virology of Hepatitis B. *Semin. Liver. Dis*. **2004**, *24*(1), 3–10.
11. Bini, E.J.; Perusmalwami, P.V. Hepatitis B Virus Infection among American Patients with Chronic HCV; Prevalence Ethnic/Racial Difference and Viral Interactions. *Hepatology* **2010**, *51*, 759–766.
12. Oh, J.K.; Shin, H.R.; Lim, M.K.; Cho, H.; Kim, D.I.,; Jee, Y.; Yun, H.; Yoo, K.Y. Multiplicative Synergistic Risk of Hepatocellular Carcinoma Development among Hepatitis B and C Co-Infected Subjects in HBV Endemic Area: A Community Based Cohort Study. *BMC Cancer* **2012**, *12*, 452.
13. Liaw, Y.F.; Chen, Y.C.; Sheen, I.S.; Chen, R.N.; Yeh, C.T.; Chu, C.M. Impact of Acute HCV Super Infection on Patients with Chronic HBV Infection. *Gastroenterology* **2004**, *126*, 1024–1029.
14. Eze, J.C.; Ibeziako, N.S.; Ikeuna, A.N.; Nwokoye, I.C.; Uleanya, N.D.; Ilechukwu, G.C. Prevalence and Risk Factors for Hepatitis C and HIV Co-Infection among Children in Enugu, Nigeria. *Afr. J. Infect. Dis*. **2014**, *8*(1): 5–8.
15. Opaleye, O.O.; Zakariyahu, T.A.; Tijani, B.A.; Bakare, A.S. HBV, HCV Co-Infection among Blood Donors in, Nigeria. *Ind. J. Pathol. Microbiol*. **2010**, *53*, 182–183.
16. Ola, S.O.; Otegbayo, J.A.; Odiabo, G.N.; Olaleye, O.D.; Olubuyide, I.O.; Summerton, C.B.; Bamgboye, E.A. Occult HBV Infection among a Cohort of Nigerian adults. *J. Infect. Dev. Ctries*. **2009**, *3*(6), 442–446.
17. National Population Commission. *National Census Sentinel*. 2007.
18. Sangeeta, P.; Meenal, S.; Buddhan, B.; Manjula, J. Prevalence and Makers of HBV, HCV and HIV in Delhi Blood Donor: A Hospital Based Study. *Jpn. J. Infect Dis*. **2007**, *60*, 389–391.
19. Yuen M.F., Wong D.K., Lee C.K., Tanaka Y., Allan J.P., et al. Transmission of HBV Infection through Blood Transfusion from Blood Donors with Occult HBV Infection. *Clin. Infect. Dis*. **2011**, *52*(5), 624–632.
20. Komatsu, H.; Inui, A.; Sogo, T.; Tateno, A.; Shimokawa, R.; Fujisawa, T. Tears from Children with Chronic HBV Infections Are Infectious Vehicles of HBV Transmission: Experimental Transmission of HBV by Tears using Mice with Chimeric Human Livers. *J.* InfectDis. **2012**, *206*, 478–485.
21. Armstrong, G.I.; Wasley, A.; Simrad, E.P.; McQuillan, G.M.; Kuhnart, W.L.; Alter, M.J. The Prevalence of HCV in the United States, 1999 through 2002. *Ann. Intern. Med*. **2006**, *144*, 705–714.
22. Ogunkunle, M.O.; Oni, A.A.; Odiabo, G.N.; Olaleye, O.D. Hepatitis B Surface Antigen (HBsAg) in Blood and Genital Secretions of Patients with Sexually Transmitted Diseases in Ibadan, Nigeria. *West. Afr. J. Med*. **2005**, *24*, 206–208.

Downloaded by [Ondokuz Mayis Universitesine] at 13:41 08 November 2014